RoadScan Technical Report

Road damage detection and human reviewed inspection

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RoadScan connects local road-image inference to saved evidence, human correction, review status, repair-priority scoring and optional nearby-report comparison. Its principal contribution is a traceable inspection workflow, not an autonomous pavement-safety decision.

Abstract

The application uses a React and TypeScript interface with a FastAPI service, local model inference and persistent report storage. It accepts still images, sampled live frames and uploaded video. A reviewer can correct damage classes and boxes, compare two or three detectors, save an explicit decision and approve a checked example for a separate training and evaluation cycle. The four target road-damage classes are longitudinal cracks, transverse cracks, alligator cracks and potholes.

The current inventory contains **22 independently SHA256-verified weight identities**, including 19 road detectors and three support assets. The supplied Main YOLO26s result is **63.04% mAP50 and 33.01% mAP50–95** on a team-defined labeled RDD holdout. These historical scores have not been independently reproduced and are affected by reported split leakage. The 343-frame road benchmark was reused for selection and comparison; it is not an untouched independent field test.

Historical experiments compare classification, detector size, augmentation, country transfer and simulated duplicate retrieval. They support the product design but do not establish a universal best model or local field readiness. The mandatory 40–80-photo city test set, real-pair retrieval evaluation, representative failure examples and end-to-end acceptance remain incomplete. Recorded software checks support bounded implementation behavior, not those missing measurements.

Technical snapshot

Main report pages 1–15; technical appendices and references pages 16–25. Source revision 58e7c3e494b5e1c5f2ca4d431b24d3f099003205. Historical experiment results, current source behavior, trainer statements and reported software verification are distinguished throughout. References and evidence identifiers appear on pages 24–25.

1 Problem users and success criteria

A road-inspection team needs to record visible damage, inspect the evidence, decide which reports deserve attention and avoid treating repeated observations as unrelated incidents. A classifier can say that damage may be present, but it cannot show which part of the image triggered that decision. RoadScan therefore uses object detection as the primary task and couples it to an explicit review workflow.

Users and decisions

The intended users are a maintenance supervisor deciding what to inspect or repair first, an inspection operator uploading imagery, and a citizen submitting a report without having a distinct defect ignored as a duplicate. An operator uploads or captures an image. A reviewer checks the whole image, edits boxes and labels, adds notes and saves a status. An administrator controls sensitive operations such as dataset export and account administration. The system supports a person's inspection decision; it does not measure pavement depth, certify structural safety, authorize road closures or autonomously dispatch repairs.

Class	Meaning	Role in RoadScan
D00	Longitudinal crack	Target damage class
D10	Transverse crack	Target damage class
D20	Alligator crack	Target damage class
D40	Pothole	Target damage class
OTHER	Other surface damage	Additional class in some checkpoints and reviewed exports

The four target classes follow the RDD/CRDDC convention. OTHER is not a fifth challenge class and is excluded from the supplied YOLO26 four-damage-class mean. Its presence in a checkpoint is a class-map fact, not permission to silently reinterpret class indices. [R1, R3, R4, R22]

Success criteria

Capability	Evidence needed for a strong claim
Recognition	Per-class detection metrics on a frozen, independent set with reviewable reference boxes
Review and traceability	Correct coordinates, controlled edits, model identity, status history and stale-write rejection
Repeat matching	Genuine same-damage and different-damage pairs with held-out calibration
Priority assistance	Transparent scoring plus measured ranking sensitivity and expert validation
Reproducibility	Exact source, model hashes, runtime, data membership, evaluation protocol and acceptance record

Boundaries

RoadScan is a local web application. It does not claim drone control, measured engineering severity, live government dispatch or autonomous repair decisions. Detailed media and map boundaries appear in the technical appendices. [R2, R6, R10]

2 Data card and split integrity

The project uses several data streams with different purposes and provenance. RDD supplies road images and boxes; the original baseline uses country separation; supplied YOLO26 experiments use a team-defined labeled holdout; later road frames were reused for development. These scopes must not be merged into one supposedly independent test. [R3, R4, R13, R22]

Source and size	Fields and use	Quality and rights status
Original baseline 8,000 train; 1,000 Czech validation; 1,000 US test	Images, country and four-class bounding boxes. Training: 6,200 India plus 1,800 China MotorBike.	Country-separated test; exact-byte checks do not rule out near-duplicate scenes.
Supplied RDD narrative 26,869 train; 5,758 validation; 5,758 labeled holdout	Damage boxes; some checkpoints add OTHER. Main adds about 14,000 external images, about 41,000 train in total.	Counts are reported, not independently recounted. Leakage reported: 2.2% validation and 2.4% holdout.
Main external sources Rome, HRP4K, SVRDD and web	Additional road imagery. Exact membership and source-by-source counts are absent.	Raw manifests and complete per-source redistribution/permission records are not established.
Road frame package 923 total; 580 train; 343 selection	Five-class YOLO labels and video-prefix/frame identities. 1,149 boxes; 461 empty label files.	Seven of eight groups cross the split; selection reused for checkpoint/model/confidence choices.
Required local set 40–80 team-collected city photos	Safely captured, labeled YOLO images; TEST ONLY and never used for training.	No qualifying untouched set established. The 343 selection frames do not fulfill this requirement.

License permission and data fields

RDD is obtained from the organizers' release, but public availability is not a blanket license for every derivative or external image. Preserve the exact dataset version, applicable terms and attribution; source-by-source license and image-use records remain a completion item. Raw imagery must not be republished on the strength of model-release permission. A local collection needs permission records and faces/plates blurred before display or sharing.

The data card for each image should bind its image/label hash, dimensions, class and original-coordinate box, country, recording/route group, split and permitted use. Report location source separately; RDD has no GPS. Review status, model identity and annotation revision belong to application provenance, not ground-truth accuracy.

Why group isolation matters

Split by country or independent recording/route groups and audit near duplicates before freezing the test. No exact duplicate bytes does not rule out adjacent frames or shared scenes. The 343 selection set has 319 crack boxes versus 31 pothole boxes, plus 42 OTHER: overall scores can hide this imbalance. A later peer record reports 26,868 RDD training images versus the older 26,869 narrative; without hash-bound manifests this discrepancy remains unresolved. Original baseline and road audit details remain in the source records. [R23]

3 System architecture and data flow

The browser presents the bilingual workflow and submits explicit user actions. FastAPI validates requests and coordinates local inference, review, model management and export. SQLAlchemy stores report and history records; the normal local configuration uses SQLite. DINOv2 embeds damage crops and FAISS searches open reports inside a GPS radius; matching remains advisory and uncalibrated. Images, models and feature caches are separate from source. [R2, R5, R7, R15]

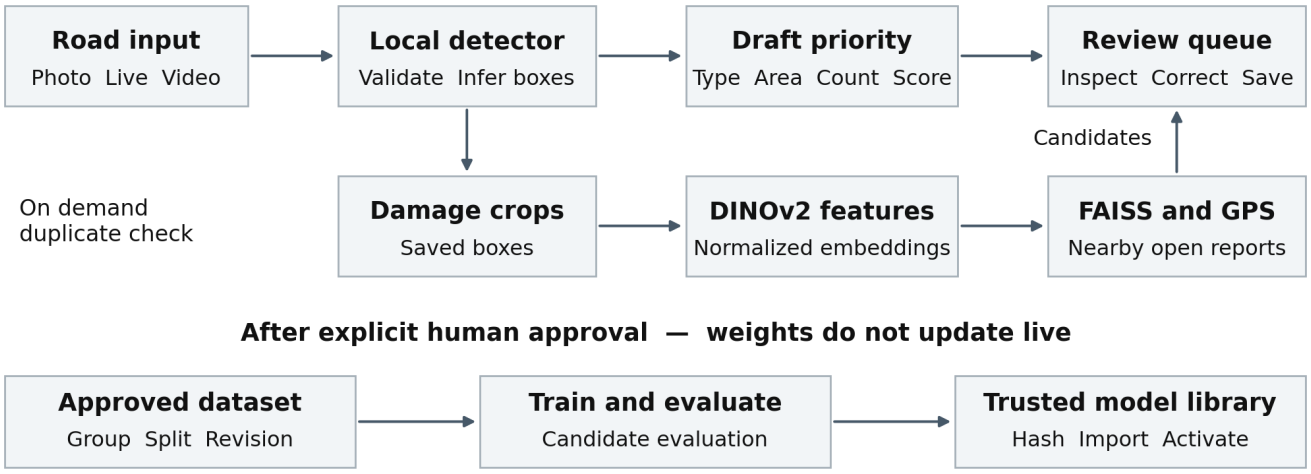


Figure 1. Detection and draft priority feed human review. Optional crop retrieval suggests duplicates; the lower row is a separate improvement cycle, not live learning.

Persistent evidence and temporary analysis

A saved report retains an image identity, the image coordinate system, predictions or reviewed annotations, inference state, model version, status and location provenance when supplied. History records track status changes, while audit events record actions and annotation changes. A review revision is computed from the report's persisted state so stale clients cannot silently overwrite a newer decision.

Temporary recognition and video analysis return predictions without automatically creating reports. This separation prevents every sampled frame from becoming a permanent incident. A user must deliberately promote evidence into the saved-report workflow. The same physical defect may appear in many analyzed frames; frame analysis alone does not count unique road defects.

Trust boundaries

Input checks protect image size, supported formats and annotation coordinates. Local detectors require the expected SHA256 and supported class mappings. Process-local locks serialize heavy work; a busy state is returned rather than allowing overlapping inference to corrupt runtime state. Model voting uses short-lived sequential workers, which release tensor memory on exit but are not a security sandbox for hostile checkpoints.

The normal Compose service binds to the local loopback address and persists data in a named volume, with a read-only external model mount. These defaults support a local workstation deployment; they do not establish an internet-hardened hosted service. Database migration utilities and other adapters should not be confused with evidence that a production multi-user geospatial deployment has been accepted. [R2, R5, R7, R8]

4 Baseline main approach and operating thresholds

RoadScan uses detection because the reviewer needs a location to inspect and correct, not only an image-level damage label. The required ResNet18 multi-label classifier is the E1 baseline; the separately trained YOLO11n road detector is an earlier detection baseline. These are distinct assets and tasks. Segmentation could describe pixel-level extent, but no segmentation evaluation is claimed. [R3, R4, R22]

Baseline and main choices

The original YOLO11n baseline provides a small, reproducible country-separated reference. Its reserved US test is 25.05% mAP50 and 10.96% mAP50–95. Main YOLO26s at 1024 pixels is the documented detailed-photo default because the supplied detector comparison supports a useful accuracy/size tradeoff and it integrates with review. Compact YOLO26n at 640 is a smaller preview alternative. Neither recommendation proves the best model for Saudi roads or a particular device.

Main's supplied labeled RDD holdout result is **63.04% mAP50 and 33.01% mAP50–95**, averaged over D00, D10, D20 and D40, not OTHER. The old US baseline test and the Main RDD holdout are different evaluations and cannot establish a fair improvement magnitude. Main's raw split manifests are missing and reported leakage may inflate performance.

Damage class	Main AP50	Validation threshold
D00 Longitudinal crack	63.55%	0.173
D10 Transverse crack	62.52%	0.166
D20 Alligator crack	69.90%	0.195
D40 Pothole	56.19%	0.056

The confidence profile is bound to the exact Main checkpoint hash. Thresholds were selected at 1024 pixels: F1 for cracks and F2 for potholes, favoring recall for missed-pothole cost. OTHER is excluded. Reduced live input has not been recalibrated; a low pothole threshold still requires human review. [R3, R5]

Metrics and model identity

IoU measures overlap between a predicted and reference box. AP summarizes the precision–recall curve; mAP averages across the declared classes. mAP50 uses IoU 0.50 and mAP50–95 averages 0.50 through 0.95. Neither is the percentage of images correctly classified. The stricter result exposes harder localization; pothole AP50 is the weakest supplied Main class result. [R1, R14]

The catalog preserves 22 verified weight identities: 19 detectors, one classifier, one COCO initializer and one embedding backbone. Four-class order ends in D40; supplied five-class order places OTHER at index 3 and D40 at 4. Imports map class names explicitly. Two additional peer-reported weights remain outside the verified count; full identities and publication scope are in Appendices B and C.

5 Experiments E1 and E2

E1 Image presence versus localization

E1 asks whether each damage type is present in an image. ResNet18 produces four image-level scores; detector boxes are converted to image-presence predictions for the comparison. The reported evaluation uses 5,758 images and image-presence thresholds, which differ from the app's box-detection thresholds. [R3]

System	Any damage F1	Image accuracy
ResNet18 classifier	0.893	0.869
Main YOLO26s detector	0.929	0.913

Class F1	D00	D10	D20	D40
ResNet18	0.790	0.751	0.805	0.639
Main YOLO26s	0.861	0.805	0.859	0.762

The detector is stronger on all reported class F1 values and additionally supplies localization for human correction. This supports using detection in the product. It does not make the classifier's F1 interchangeable with detector mAP, and the comparison still requires a frozen manifest and reproducible preprocessing to become independently verified.

E2 Size and operational tradeoffs

At 640-pixel input, the supplied small model scores 59.43% mAP50 versus 55.99% for nano, a 3.44-percentage-point difference. A separate local diagnostic favors nano on model time. These answer different questions: historical detection quality on a stated split and short-run processing time in a stated environment.

24 October diagnostic	Median model time
Nano 640	57.89 ms
Small 640	122.38 ms
Main Balanced 640	124.74 ms
Main Detail 1024	298.94 ms

The timing diagnostic used only three warm requests on one synthetic 960 × 640 image with Docker limited to two CPUs and 4 GiB. It excludes upload, decode, browser rendering and camera acquisition. It is not representative field latency or camera FPS, and it should not be combined with later preflight timings that used a different budget and included decode/preprocessing.

Operating point rather than a winner

Use Main as the documented detailed-photo default and consider Compact when preview responsiveness matters. Before choosing for a device, measure acquisition-to-result latency, warm and cold behavior, memory and relevant class performance on that device. Smaller inputs can change small-defect recall and threshold behavior. Model name, checkpoint file size and synthetic timing are insufficient to establish the best field choice.

6 Experiments E3 and E4

E3 Augmentation ablation

The standard 640-pixel small model exceeds the no-augmentation run by 12.50 mAP50 points. Vertical flipping does not improve the standard run. The comparison supports the chosen policy under this experiment. Horizontal changes may reflect lane/view variation, while vertical flips invert road geometry and camera perspective and are physically implausible for ordinary road capture; more augmentation is not automatically better. [R3]

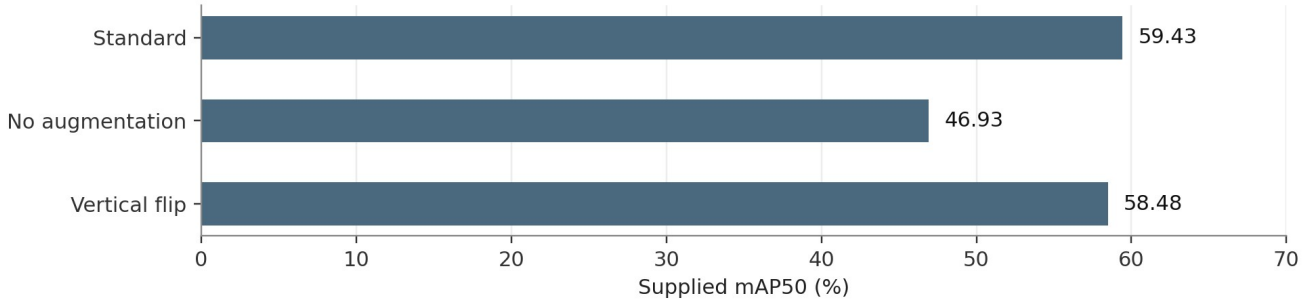


Figure 2. Supplied E3 comparison on the team-defined RDD holdout. Input size and model size are held at the stated small 640 configuration.

The cleaned-label run reports 60.05% holdout mAP50 and 30.73% mAP50–95. Its validation mAP50–95 is 0.3403 versus 0.3421 for the original-label comparison, so Main retained the original labels. A favorable holdout number must not be substituted for the validation-based selection decision. Complete cleaning rules, membership and labeling quality still need reproducible provenance.

E4 Held out country

India evaluation condition	Supplied mAP50
India excluded from comparison-model training	8.86%
India included in comparison-model training	39.25%

The decline illustrates domain shift under the supplied experiment: appearance, surface, capture angle and surrounding context can affect transfer even when damage labels have the same names. It does not forecast a Saudi score or quantify the contribution of any one visual factor. A local independent test is needed rather than assuming international benchmark performance transfers unchanged.

Required city evaluation remains incomplete

E4 requires the held-out-country comparison and a separate 40–80-photo local test collected safely by the team, labeled and never used for training. No qualifying untouched local set is established. The later 343-frame road set was used for selection and cannot substitute. Domain-shift performance must be reported separately by class, including which defects are missed, rather than inferred from the India result. [R22]

Main also has supplied external mAP50/mAP50–95 pairs: HRP4K 53.00/31.04%, Rome 35.07/14.49%, web 82.13/51.93% and SVRDD 67.63/38.85%. These are distinct measurements. Some source names also occur in training; absent membership manifests, the names alone do not establish independence.

7 Experiment E5 and duplicate retrieval

Repeated views of one defect can inflate an inspection queue, while merging nearby but distinct defects can hide necessary work. RoadScan therefore presents duplicate candidates for human inspection rather than merging automatically. The historical E5 experiment and the current crop-retrieval implementation have different pipelines and must be interpreted separately. [R3, R15]

E5 Supplied simulated observations

Supplied method	F0.5
pHash with GPS	0.851
ResNet50 with GPS	0.975
DINOv2 with GPS	0.980
DINOv2 without GPS	0.796

DINOv2 with GPS reports precision 0.983, hard-repeat recovery 0.951 and a new-report false-merge rate of 0.032; without GPS, the reported false-merge rate is 0.137. The simulation supports testing geographic candidate filtering, but it is not measured duplicate accuracy of the current application and does not establish a complete real-pair Recall@5 or ROC study.

Current damage crop search

The optional matcher embeds saved damage crops with the official pretrained DINOv2 ViT-S/14 backbone. It uses 384-dimensional features, the declared crop/518-pixel/ImageNet preprocessing, L2 normalization and exact FAISS inner-product search, equivalent to cosine similarity after normalization. The cache key includes checkpoint identity, preprocessing, image content and crop coordinates.

Candidates must be open reports within the default 50-metre radius, with boxes and compatible location source. The query itself and resolved/duplicate reports are excluded. Manual and reported device-GPS sources can compare with each other; simulated positions compare only with simulated. Unknown sources are excluded. The source bounds a search at 100 nearby reports and 20 boxes per report, then returns up to five highest-scoring report matches. A first search can be slower while crops are embedded.

Thresholds and the lightweight alternative

FAISS end-to-end execution on the final head remains unverified. The current DINOv2 threshold 0.80 is experimental, not 80% accuracy. The teammate’s E5 threshold 0.6529684 must not be transferred without demonstrating equivalent preprocessing and calibration. The separate lightweight full-image path is an 8 × 8 **difference hash**, with an unvalidated similarity threshold of 0.85; despite a legacy function name, it is not the E5 DCT pHash method.

Required calibration

The brief requires team-built real same-damage/different-damage pairs and comparison of perceptual hash, ImageNet ResNet features and DINOv2. That real-pair study remains incomplete. Collect pairs across light, viewpoint and distance. Keep recording/site groups disjoint, select the threshold on validation only and report test precision, recall, F0.5, false merges and Recall@1/5 and an ROC curve with denominators. [R22] GPS provenance and error matter: a radius can remove false candidates but also exclude a true repeat when its coordinates are wrong. Similarity remains advisory until that evidence exists.

8 Priority scoring and sensitivity

RoadScan computes an interpretable **draft priority**, not measured engineering severity. It combines damage type, visible box extent, count and confidence into a score from zero to 100. A report without boxes has no score; it is not classified as safe or zero risk. [R16]

$$S = 100 \times (0.40\,T + 0.25\,A + 0.20\,N + 0.15\,C)$$

Term	Definition	Weight
T Damage type	Maximum class weight: D00 0.55, D10 0.45, D20 0.85, D40 1.00, OTHER 0.55	40%
A Relative area	Largest box area divided by the whole image area	25%
N Count	Minimum of box count divided by five and one	20%
C Confidence	Mean score stored with the detections or reviewed annotations	15%

Why the formula needs interpretation

Type has the largest coefficient and prioritizes potholes and alligator cracking under the chosen design assumptions. Area and count add visible extent, and confidence contributes to the remaining term. These weights are understandable and inspectable, but their transparency does not make them an official maintenance policy or validated engineering scale.

Perspective, camera distance and framing change the fraction of the image covered by a defect. Box area cannot measure depth, road width, traffic exposure or load-bearing risk. Confidence is a model score rather than physical damage magnitude; after human annotation it reflects the saved annotation value and should not be described as calibrated certainty.

Worked illustration

For one pothole box covering 10% of an image with a stored score of 0.80, T is 1.00, A is 0.10, N is 0.20 and C is 0.80. The formula gives 58.50. This is a mathematical illustration, not a measured road condition, a repair instruction or an empirical threshold for urgent work. Changing the camera crop can change that number without changing the pavement.

Sensitivity tools and missing validation

The per-report tool changes each weight by ±20% and renormalizes the full vector, producing eight scenarios. The dataset tool applies those scenarios to the supplied reports, orders by score and report ID for deterministic ties, and measures maximum rank shift and top-k overlap with k at most 50. Equal scores are not distinct severity judgments.

These tools establish a mechanism for inspecting sensitivity. The brief's empirical ranking-sensitivity requirement is not completed by implementing the tool: representative-road ranking results and expert-agreement validation are not established. [R22] A useful next study would compare rankings with independent reviewer priorities, report unstable cases and separate the influence of viewpoint, annotation changes and weight choice. The deployed formula should not be silently replaced by a different narrative formula or nonlinear transform.

9 Application workflow media and artifact loading

The local application connects input, boxes, review, draft priority and duplicate candidates. The core flow is deliberate: upload an image, inspect the whole result, correct annotations, save a decision, then optionally approve a training example. It does not turn a prediction or majority vote into ground truth. [R2, R5, R7, R8]

Input or action	What happens and what does not
Photo upload	Default 50 MiB and 25 megapixels. The stored JPEG strips EXIF; source bytes and embedded GPS are not retained. A report is saved.
Temporary recognition	POST /api/recognize accepts a 2 MiB / 2 megapixel frame and returns state, model identity and boxes without a report.
Live or uploaded video	Live frames are serially sampled. Video offers every frame or 10/5/2/1 per second. Temporal summaries are not unique defects or locations; no automatic reports/GPS.
Human review and voting	Revision-checked edits and audit history; two or three distinct detectors give an advisory second opinion. Failures withhold consensus; a minority prediction may be chosen.
Approved dataset export	Confirmed review plus rights, group and split approval creates a local YOLO ZIP. Training, evaluation and activation remain separate.

Model loading and result interpretation

Import a trusted compatible checkpoint with its SHA256 and metadata in Settings, then choose Use model and confirm Active. Import alone does not select it; activation checks loading and a synthetic forward pass while failure preserves the prior model. Restricted loading is not a sandbox. A completed empty result means no boxes at that operating point, not a safe road; missing, busy and failed states remain distinct.

Location and navigation

Map & results is a page-fit coordinate diagram with independent search, status and paging. Recorded/manual or reported device-GPS positions are separate from simulation and unusable/unknown locations. Older servers show an explicit loaded-page-only fallback. The app does not extract GPS from photos/video or edit saved coordinates. Google Maps is an explicit external coordinate link, not an embedded basemap. [R9]

Website guide is curated English/Arabic help, not an LLM. It reads no report contents and navigates only after a user click; it cannot upload, infer, save, activate or change permissions. Guards protect unsaved review and child workflows. Detailed video, map and review contracts are in Appendices E–G. [R10]

Latency and demonstration scope

The E2 table reports narrow historical model-only CPU timing, not acquisition-to-result latency. Representative end-to-end measurements and physical-camera acceptance remain pending. The required demonstration must use a panel-chosen unseen photo, show intermediate boxes/priority, then a second view and duplicate suggestion with queue behavior. Screenshots are not a live demo; a recording is only a reduced-mark fallback after live failure. [R22]

10 Held out and team built evaluation

Evaluation must distinguish data never used for decisions from development data reused for selection. Software tests and successful model loading answer a different question from measured damage recognition. The official brief requires both held-out and team-built evaluation with per-class evidence. [R22]

Evidence set	Available result	Interpretation
Original baseline US test	1,000 reserved images; 25.05% mAP50, 10.96% mAP50–95	Country-separated baseline evidence, not unseen-data proof for imported models.
Team labeled RDD holdout	Main 63.04% / 33.01%; supplied per-class AP50	Historical, not independently reproduced; reported 2.4% near-duplicate leakage.
India-only experiment	8.86% mAP50 when excluded versus 39.25% when included	Supplied E4 country-shift comparison, not a forecast for Saudi roads.
Required local test	40–80 team-collected city photos, test only	No qualifying completed result; selection frames do not fulfill the requirement.

The road package is a selection benchmark

The later package has 923 frames: 580 reconstructed training and 343 selection, with 392 selection boxes. Seven of eight filename video groups cross the boundary and 69 adjacent pairs cross splits. There are no exact image-byte duplicates across the split, but that does not establish scene independence. The same selection frames were used for checkpoint, confidence and model comparison. [R13]

Historical result on the 343 selection frames	mAP50	mAP50–95
Main	39.22%	26.63%
Road v1	46.74%	31.26%
Road A	46.80%	28.98%
Road v2b	48.61%	29.37%

Supplied big_test_4_models.json; four damage classes, OTHER excluded. Results were not rerun. P/R/F1 and false alarms compare each model at its own best-F1 confidence; separate Kaggle tables use a different protocol.

Road v1’s recorded F1 0.53724 and 10.50 false alarms per 100 frames are therefore development results. The evaluator is absent and the historical v1 export-selection explanation remains unresolved. None of these numbers proves a statistically reliable winner or an independent local-test improvement.

Evidence still required

Freeze a new permitted city set with image/label hashes and recording/route groups outside all training and selection. Report per-class AP, precision, recall, confusion matrix and F1-versus-confidence, with counts and representative errors. Complete real-pair Recall@1/5, threshold precision/recall and ROC for duplicate retrieval. No full fair common-set detector evaluation, qualifying city test or complete real-pair study is established. Appendices A–D preserve the audit and model provenance.

11 Error analysis and responsible AI

Predictions require human review. An empty output, low priority or majority agreement cannot certify safety. RoadScan does not replace qualified pavement assessment.

Failure category	Evidence to collect
Missed defect	A reference box with no accepted prediction, including class and capture condition
False alarm	A marking, shadow, repair patch or surface texture predicted as damage
Class confusion	Reference and predicted class for the same localized defect
Poor localization	A matched defect whose box misses extent or includes excessive context
Repeat matching error	A missed genuine repeat or a false merge between nearby different defects
Capture degradation	Paired observations across blur, glare, low light, distance and viewpoint

Required real failure examples, counts and a reviewed gallery remain pending; these are example categories to collect, not observed frequencies. Save a representative error gallery with model hash, input size, thresholds, reference annotations, denominators and reviewer rationale. Do not select only attractive predictions or infer a specific failure cause from a lower aggregate score.

Privacy and location

Images can contain faces, number plates and identifying surroundings. JPEG re-encoding strips EXIF, but it does not remove visible personal information. Automatic face/plate redaction is not implemented. The brief requires faces and plates blurred in team photos and anything displayed. Define permission, access and retention before collection or sharing. [R22]

Stored coordinates can reveal sensitive routes or locations. Manual and reported device-GPS labels communicate provenance but do not verify truth. The external Google Maps action shares coordinates only when the user clicks; it should not be presented as an offline operation. Training exports omit GPS and notes but still contain images and reviewer identifiers, so they remain private workspace data.

Security and operational limits

Access controls, validation, revision checks and hashes reduce specific risks, not internet-facing security assurance. Keep tokens/backups private and import only trusted checkpoints. Worker isolation does not make hostile model content safe.

Coverage and fairness

Road surfaces, neighborhoods, lighting, devices and capture angles may be unevenly represented. Citizen under-reporting can under-serve neighborhoods even with an equally accurate detector. Measure subgroup performance only with meaningful sample sizes and report coverage gaps. Repeated frames from one route do not substitute for diverse independent cases. Priority weights may favor visible size or a damage category without capturing traffic exposure, so any operational ranking needs expert interpretation and documented policy.

One documented implementation failure was an RT-DETR box crossing the image border and causing whole-frame rejection. The adapter now clips finite coordinates and drops zero-area boxes while preserving strict human-label validation. This functional regression example is not a substitute for real false-positive, missed-defect and class-confusion examples. [R6]

12 Reproducible artifact and improvement cycle

The public repository and pinned revision now provide a concrete source identity. Reproducing inference still requires a trusted detector, its full hash and the documented environment; cloning source alone does not provide every model or reproduce historical training. [R4, R5, R11, R18]

A controlled inference path

- Obtain the pinned source and use the documented Windows/Docker setup path. The Docker build pins base/dependency constraints and uses npm ci with the frontend lockfile; a loose pip installation is not equivalent.
- Enable detector support when the intended setup needs it. Download an eligible trusted weight with its metadata and notices, and verify the complete SHA256 and byte count before import.
- In Settings, import the checkpoint or prepared model package, then choose Use model and confirm Active. Import alone does not activate it. Failed activation must leave the previous detector available.
- Test temporary recognition and a saved-report review separately. Verify the returned state, model version and persistence behavior, then restart and confirm the selected model and stored data remain.

Model and runtime safeguards

The library rejects unsupported class maps and verifies a synthetic forward pass before a model swap. It supports trusted .pt files and bounded model ZIPs; restricted deserialization reduces unsupported serialized-code exposure but is not a sandbox. Source settings use one worker for process-local locks. Credentials, report images, local configuration and private rollback snapshots must stay outside public source packages.

Named storage volumes and read-only model mounts make data independent of a disposable service container. Preserve the existing Compose project name, .env and named volume; do not use docker compose down -v as an upgrade step. Before an authorized replacement, stop and back up consistently, preserve configuration and model identities, and compare database/file hashes and counts afterward. A successful restart is weaker evidence than a source-built image recreation with preserved state.

Historical runtime evidence versus current source

The 31 October record documents a source-built r5 image, 116 backend tests in that image and one synthetic Main recognition request with persisted=false. The production 19-report store and stopped nine-report demo store retained their recorded hashes and configuration. Those are preservation and detector-operation checks, not an evaluation dataset or benchmark. [R19]

The 1 November main receipt is separate and does not establish deployment. A clean public-clone, downloaded-weight, fresh-process inference acceptance remains incomplete; neither synthetic smoke nor a published download fulfills that requirement. [R20, R22]

Training reproduction remains incomplete

The original baseline has recovered source and configuration evidence, but teammate raw manifests, some historical initializers, selection explanations and evaluator scripts remain incomplete. Review corrections can be approved into a grouped train/validation/test export, but weights change only through a separate train/evaluate/import cycle. Exact artifact identity does not reproduce missing training history or prove test independence. [R17]

13 Requirement status and verification

The application has a documented software baseline, but project completion also depends on empirical evaluation and demonstration. The following matrix maps the official Track A requirements to the available evidence; a tool or historical result is not silently promoted into a completed independent study. [R22]

Requirement	Available evidence	Remaining completion condition
E1 classifier versus detector	5,758-image image-presence table with class F1	Reproduce preprocessing/threshold protocol from frozen manifests.
E2 size and CPU tradeoff	Small/nano scores and three-request CPU diagnostic	Representative end-to-end and device latency, with sample counts/tails.
E3 augmentation	Standard/no-augmentation/vertical-flip comparison	Preserve comparable recipes and repair split leakage before stronger claims.
E4 country plus local test	India-held-out result; Main class AP	40–80 safely collected, labeled city photos kept test-only, with class results.
E5 real duplicate pairs	Three-method simulated comparison; crop/FAISS implementation	Real pairs, validation-only threshold, Recall@1/5, precision/recall and ROC.
Priority sensitivity	Formula, eight perturbations and ranking-analysis tool	Representative ranking-change results and interpretation.
Error examples and responsible AI	Defined failure categories and privacy/coverage risks	Real annotated failures/counts; displayed faces and plates blurred.
Artifact and live demo	Public source/weights and bounded integration checks	Clean-clone fresh-process acceptance; panel-chosen unseen-input flow.
Team contribution record	Owner reports backend/models/frontend allocation	Detailed task log and meaningful repository contributions remain to be confirmed.

Current source verification is bounded

The 1 November exact-main receipt records 257 backend passes with three environment skips, 42 frontend passes, type checking/build and isolated API/browser checks. Main is 58e7c3e494b5e1c5f2ca4d431b24d3f099003205; full English/Arabic guards belong to its identical tested tree. The synthetic integration used 28 reports: 24 recorded, two simulated and two without usable location, with recorded pages of 20 plus 4. [R20, R21]

The candidate also records Vite 6.4.5, a dependency audit with zero findings and 126 checked catalog links. These are local records, not CI, a comprehensive security audit, inference accuracy or deployment acceptance. The server was stopped. Historical preflights covered 14 of the then-16 detectors on 20 images; no full common-set benchmark was completed, and imported training overlap with the US set is unknown. [R4]

14 Conclusion next steps and contributions

RoadScan's established contribution is an integrated local workflow: damage predictions become persistent evidence with revision-protected human review, transparent draft priority and advisory duplicate candidates. Historical E1–E5 results support design choices, while current software checks show bounded implementation behavior. Neither establishes complete field readiness.

The central scientific limitation is evaluation independence. Reported RDD leakage and the reused 343-frame road selection set prevent stronger generalization claims. The required 40–80-photo city test, real duplicate-pair study and representative failure gallery remain incomplete. The honest conclusion is a working, inspectable prototype with useful experiments and explicit remaining evidence, not a fully validated municipal repair system.

Next steps in priority order

- **Protect a real city test.** Safely collect and label 40–80 permitted photos, blur visible faces/plates, freeze image/label hashes and keep them test-only. Use separate local development data if fine-tuning is attempted; do not train on the required test set.
- **Complete evaluation rather than recycle selection.** Audit near duplicates and route/session groups, freeze model and thresholds before scoring, and report per-class errors. Build real same/different-damage pairs for the three retrieval methods, choose thresholds on validation, then report Recall@1/5 and ROC on disjoint groups.
- **Validate the operational decisions.** Run ranking sensitivity on representative reports, measure cold/warm end-to-end CPU latency and verify the clean-clone artifact path. Retain Main and all unique model identities for rollback and reproducibility.
- **Rehearse the required demonstration.** Use an unseen panel-chosen photo, show boxes and draft priority, compare a second view, inspect the duplicate score and demonstrate queue behavior. Screenshots are not a demo; a backup recording is only a contingency. [R22]

Contribution statement

Team member	Owner reported responsibility
Feras Alyamani	Backend
Ahmed Salamah	Models
Ahmed Alhejaili	Frontend

The owner confirmed these high-level responsibilities for this report. They do not establish specific subtasks, authorship of every script or meaningful commit counts; a detailed contribution log and repository history should support those claims. Suggested Member 1/2/3 roles in the brief are examples, not actual assignments. AI-assisted model/source records are retained as provenance rather than recast as independently verified human work. Every member must understand and explain the complete system.

Reading map: pages 1–15 are the main report. Appendix A audits the road data; B–D preserve model identities and provenance; E–H document media, review, map and learning contracts; references occupy pages 24–25. No new model training, field collection or accuracy run is represented by this report rewrite.

Appendix A Road benchmark identity audit

The later road package contains 923 labeled frames. A read-only identity audit reconstructed its split and established that the 343-frame subset was used for checkpoint selection, confidence comparison and model comparison. It is useful as a development benchmark, but it is not an untouched independent test. [R13]

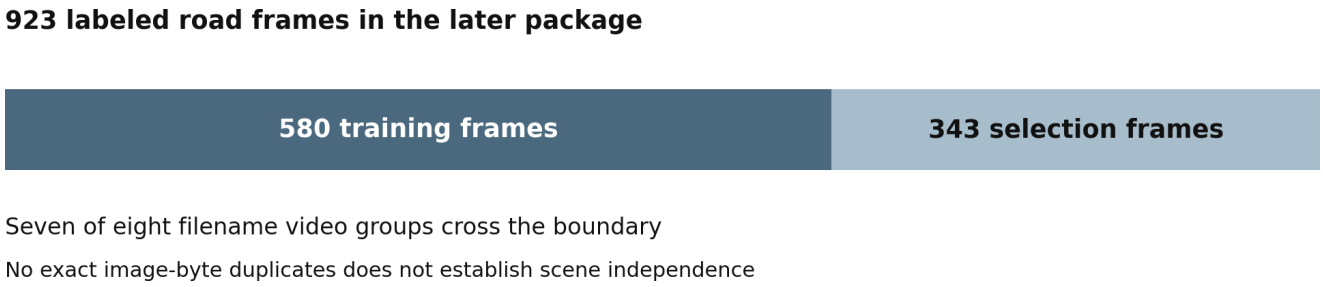


Figure 3. Reconstructed membership of the later road-frame package. The partition is an exact manifest reconstruction, not evidence of independent capture groups.

Partition	Frames	Boxes	Empty labels
Reconstructed training	580	757	288
Selection manifest	343	392	173
Total	923	1,149	461

The selection boxes are D00 235, D10 54, D20 30, OTHER 42 and D40 31. The training boxes are 442, 87, 85, 106 and 37 in the same order. Empty files are part of the package; their presence does not independently establish the quality of negative-label review.

Why the partition is not independent

The reconstructed rule sorts frames within filename prefixes and alternates groups of five sampled frames: three blocks for training and two for selection. Seven of eight filename video groups cross the split, with 69 consecutive-within-prefix pairs crossing the boundary. No exact image-byte duplicates cross it, but nearby frames and shared scenes can remain highly correlated.

Both A/B selection manifests list the same 343 unique frames. All 150 frames from the first selection round remain, and none of the old 240 training frames enters the new selection set. Original train/test folder names are historical and do not define current membership. Earlier manifests name files but do not establish unchanged label bytes across versions.

Consequences for interpretation

The saved fine-tune script scores the initial model and each epoch on road data and selects a checkpoint using those scores. The package also compares models at their own best-F1 confidence on the same 343 frames. Its recorded Road v1 F1 of 0.53724 and about 10.50 false alarms per 100 frames are therefore selection-benchmark results, not independent generalization estimates. The evaluator for the separate big-test output is absent.

A genuinely new test needs immutable image and label hashes, original recording/session/route groups, and provenance showing no use for training, checkpoint, threshold or recipe selection. A new folder name, a count of 343 or exclusion from gradient updates is insufficient. Original video hashes and independent route identifiers are not established in this package.

Appendix B Verified road detector inventory

The catalog has **19 road-detector identities**. The abbreviated keys below are the first 12 SHA256 characters, not interchangeable app IDs. Preserve full hashes and original filenames in the machine-readable inventory. Input size is the recorded configuration, not proof that every current workflow always uses that size. [R4, R11]

Detector	SHA256 key	Classes / px	Release route
YOLO11n road baseline	2bb146ad53a3	4 / 640	Published
RT-DETR X early E1	bda15131e5f6	4 / 640	Metadata only
RT-DETR X E50	edb35d3cf7b1	4 / 640	Metadata only
YOLO11m E86	9e2a0484d6f4	4 / 640	Metadata only
YOLOv8m E97	9a13443ad7a3	4 / 640	Metadata only
RT-DETR L RDD2022 E68	b636269a3035	5 / 800	Metadata only
RT-DETR setup E1	11602fd792d5	4 / 640	Metadata only
YOLO11s RDD2022 E85	e1937609619f	5 / 800	Metadata only
YOLOv8m RDD2022 E21	4e14d23d45da	5 / 800	Metadata only
YOLO26s standard	31b518ed36ec	5 / 640	Published
YOLO26s cleaned labels	1bb6724ee2d0	5 / 640	Published
Main YOLO26s	9f940caa9222	5 / 1024	Published
Compact YOLO26n	3be62681e3ec	5 / 640	Published
YOLO26s no augmentation	3ef405bf22a9	5 / 640	Published
YOLO26s vertical flip	63d2099e766f	5 / 640	Published
YOLO26s India excluded	fa148e7109aa	5 / 640	Published
YOLO26s Road v1	78f5e2afb82d	5 / 1024	Published
YOLO26s Road A	00794102e39e	5 / 1024	Published
YOLO26s Road v2b	0caef4521c82	5 / 1024	Published

Class maps must follow the asset

Four-class detector order is D00, D10, D20, D40. The supplied five-class order is D00, D10, D20, OTHER, D40. Consequently index 3 means pothole only in the four-class group. RoadScan maps names explicitly. The exported reviewed-dataset order is different again: D00, D10, D20, D40, OTHER; it must be remapped by class name before merging with external labels.

Near-zero early RT-DETR checkpoints remain in the catalog for traceability. A filename containing an epoch number is a provenance label, not independently verified training history. No unique checkpoint is justified for irreversible deletion by this inventory or by the non-comparable scores.

Appendix C Support assets and publication

The 19 road detectors are accompanied by three assets serving different tasks. Together they make **22 locally verified weight identities**, not 22 trained damage detectors. The catalog separates byte identity, purpose, publication state and training reproducibility. [R4, R11, R12]

Support asset	SHA256 key	Purpose and publication
ResNet18 classifier	267c1eefc78d	Four-label image presence; metadata only
Stock COCO YOLO11n	0ebbc80d4a76	Official initialization checkpoint; upstream link
DINOv2 ViT S14	b938bf1bc15c	Pretrained visual retrieval backbone; published

ResNet18 does not produce boxes. The trainer identifies torchvision ResNet18_Weights.IMAGENET1K_V1, four outputs, 20 epochs and 512-pixel input. That initializer identity is now trainer-attested; the original classifier training script is still absent and the applicable redistribution basis remains unresolved. The E5 ResNet50 comparison method is not the retained ResNet18 weight and is not an extra verified catalog asset.

Current publication scope

Twelve weights have public release assets: eleven detectors and DINOv2. One stock initializer has an official upstream link. Eight older detectors and ResNet18 remain metadata-only under individually recorded source or terms conditions. A general publication authorization does not establish every asset's applicable terms, exact historical training recipe or clean-install acceptance.

The seven RunPod YOLO26 weights and three road fine tunes were released with original bytes and adjacent source/notices. The release record documents anonymous byte-count and SHA256 verification on 31 October. The 1 November inventory pass streamed existing local files and archive entries; its remote HEAD checks confirmed availability and expected lengths, not a fresh download-and-rehash of all remote weights.

Proposed organization rather than a runtime change

- **Default demo:** Main for detailed images, with Compact as a smaller alternative. This is an operational recommendation, not a universal accuracy or current device-speed ranking.
- **Experimental and comparisons:** Road v1 stays explicitly experimental. Preserve the original baseline and the three historical 800-pixel voting comparators for reproducibility.
- **Support and research archive:** Keep DINOv2 distinct, and retain all other unique research identities with metadata, source, notices and restoration mapping.

These groups and display aliases are documentation recommendations, not evidence that installed models were moved, renamed, deleted or activated. Import does not select a model; Use model and a confirmed Active state are separate steps.

Two peer reported additions

RT-DETR L E70, key d66e7ddacc35, and a YOLOv8m 183-labelled variant, key 02ab48febc4e, are recorded separately. They were not independently fetched, hashed, loaded or published in the review. Their metadata must not inflate the verified inventory to 24. The second asset's embedded epoch is −1 and its filename alone does not prove 183 training epochs.

Appendix D Training provenance and protocols

The current weight identities are verified, while historical runtime and parent-byte claims remain scoped. Main to v1, Main to A and v1 to v2b are trainer-attested and consistent with saved filenames, but exact historical parent hashes are absent. [R11, R12, R23]

Main run settings and version evidence

A later source-backed peer receipt identifies yolo26s.pt pretrained initialization, 120 epochs and patience 120, batch 12, input 1024, optimizer auto, lr0/lrf 0.01, cosine learning rate, three warmup epochs, mosaic 1.0, close_mosaic 15, horizontal flip 0.5, vertical flip zero, AMP and seed 0. The 120-row curve selects epoch 84 with validation mAP50–95 0.3668 and mAP50 0.666. These are validation values, not test results.

The training data YAML lacks hash binding. The peer count is 26,868 RDD train images versus the older 26,869 narrative, plus approximately 14,000 external Main images in that narrative. Exposure/membership remains unresolved. Stock Ultralytics 8.4.177 is the earlier trainer attestation and matches embedded metadata; the exact historical installation is not independently proven.

Road fine tune recipe and actual training counts

The first v1 package contained 390 frames: 240 training and 150 selection. A/v2b use 580 training and 343 selection out of 923. Training mixes RDD train with repeated road frames toward 2,400 road entries; total reviewed frames are not gradient-training counts. The common recipe is five epochs, 1024 input, SGD, lr0 0.0005, lrf 0.1, freeze 10, zero warmup, mosaic 1.0, close_mosaic 2, mixup 0.1, scale 0.5, translate 0.1, horizontal flip 0.5, cosine schedule, AMP and seed 42. Reported batches are six for v1 and 12 for A/v2b.

A separate common comparison protocol

P2 RDD test with one label per box	mAP50	mAP50–95
Main	60.6%	30.5%
Road v1	60.4%	30.1%
Road A	60.1%	30.0%
Road v2b	59.9%	29.9%

Peer-reported historical execution on 5,758 RDD images. P2 uses a distinct one-label-per-box protocol; its common scoring script is absent. Do not mix with P1 Main 63.04%/33.01%, embedded validation metrics or the P3 road-selection table. Small observed gaps are not proof of statistical equivalence.

The v1 historical written selector rejects all recorded epochs; actual manual selection/export is unresolved. The later v2b rule is consistent with epoch 3 but selection reuse remains. The first v1 selection was 150 frames, not all later 343. Peer v1/v2b confidence profiles are local-only and are not current repository/app settings. Preserve these distinctions before any activation or performance claim.

Appendix E Video analysis contract

Full-video analysis is a bounded local decoder workflow, distinct from sampled live preview. The interface offers Every frame or 10, 5, 2 or 1 analyzed frame per video second. The backend processes selected frames sequentially at inference speed; it does not promise real-time processing or that every uploaded codec will decode. [R6]

What a session does

The authenticated upload creates a temporary session bound to the requesting actor and current model version. Accepted extensions cover MP4, MOV, M4V, AVI, MKV and WebM, with a media-container signature check before OpenCV/FFmpeg decoding. Advertised dimensions and frame count are checked. Every-frame mode analyzes each decoded frame, while sampled modes use timestamps and disclose a nominal-FPS fallback if timestamps are invalid or non-monotonic.

Control	Current source limit
Video upload	Default 2048 MiB; 15-minute total and 60-second idle upload deadlines
Session capacity	Two simultaneous sessions; 15-minute inactivity lifetime
Image and coverage bounds	25 megapixels per frame; 50,000 analyzed-frame bound
Temporary storage	256 MiB reserve checked during upload
Pre-inference resize	Longest side at most 960 in Balanced or 1280 in Detail

The detector may resize again to its own input size. A 4K source therefore does not mean 4K model inference. Skipped sampled frames are decoded but not converted to images for inference; failures of conversion on skipped frames are consequently outside that observed path.

Damage sequences are review aids

The interface groups detections across analyzed-frame indices. A gap of at most two can keep a group together, allowing one analyzed frame with no detections between positive frames. It unions the detected classes for the sequence and retains the highest-confidence frame as the representative. It lists at most 200 sequences; later sequences are not counted or listed after that cap. These are temporal groups, not tracks that prove physical identity, unique defects or separate geographic locations.

Selecting a representative frame starts a new saved-report analysis of that frame. The union of sequence-wide classes is not copied into the report as if every class were visible there. No reports or GPS positions are automatically created from the uploaded video.

Completion cancellation and evidence limits

Decoder end is called complete only when positive processing and advertised coverage agree; unknown or mismatched metadata produce an explicit coverage-unverified result. A model change ends the session. Cancellation releases temporary files and decoder resources; an in-flight inference cannot be interrupted safely, but cleanup prevents further decoding afterward. Source inspection and recorded synthetic checks support these control paths, not acceptance of long high-resolution clips, physical cameras or drone footage.

Appendix F Review and model voting

Review is the point where the model's proposal becomes a human decision. The reviewer should inspect the whole image, remove false boxes, add missed defects and correct class labels before choosing a status. Notes, boxes and status are saved deliberately; a confident prediction or majority vote is not automatically confirmed ground truth. [R2, R7, R8]

Revision protected changes

The API validates the supplied revision before an update and includes the persisted revision inputs in the database write condition. If another request changes the report meanwhile, the write is rejected with a conflict and the reviewer must reload. Saved annotation changes recalculate the draft priority, record the prior predictions in audit details and mark the output as a manual review rather than continuing to present it as unchanged model inference.

State or action	Meaning
Needs Review	Evidence is saved but the inspection decision is not yet confirmed.
Confirmed	A reviewer has checked the report; eligible for a separate training approval.
Assigned or Resolved	Workflow states; these do not certify structural safety or dispatch work externally.
Duplicate	Human workflow classification, not an automatic merge from a similarity score.
Training approval	Separate, revision-bound permission and split assignment; not a weight update.

Voting protocol

A saved image can be compared using two or three distinct checkpoint hashes. Models run sequentially in isolated worker lifetimes with a 90-second worker timeout. The active detector selection and saved boxes remain unchanged unless the user explicitly adopts a proposal and saves the review. Live inference can report busy while voting holds the runtime.

The fusion matches same-class boxes to a fixed seed at IoU 0.50 and allows one vote per checkpoint in a cluster. Two models require both; three eligible models require two. A model that does not support a class abstains, but a class still needs at least two supporting models. Coordinates are confidence-weighted averages and the displayed score is the mean detector score, not a calibrated probability that the defect is real.

If any selected model fails or times out, consensus is withheld rather than weakening the quorum. Agreement and disagreement remain visible, and the reviewer may select an individual minority prediction. Ensembles can suppress real damage or reinforce shared errors; no measured ensemble improvement is established here.

Unsaved work

Screen navigation guards protect unsaved review edits and child workflows such as voting, training fields and video analysis. Cancel preserves the mounted work; Continue intentionally discards local state or cancels pending activity. The guard is conservative and may ask even when a selected report or child-workflow screen is idle. It does not turn ordinary navigation into permission to save or activate a model. [R10]

Appendix G Map and bilingual guide

Map & results finds a saved report by its recorded location and opens its image for review. It is a coordinate diagram, not a street basemap, route planner, distance scale, geographic survey or severity map. Positions fit the current page, so changing a page or filter can move pins on screen without any underlying coordinate change. [R9, R10]

Location group	Displayed meaning
Recorded locations	Valid manual or reported device-GPS coordinates; neither provenance is independently verified.
No usable location	Missing or invalid coordinates, or unspecified/unknown provenance. Reports stay available in the list.
Simulated locations	Explicit demonstration coordinates, kept separate from recorded positions.

Paging filters and evidence inspection

The map owns its search, status filter and pagination independently of Review queue. The additive API accepts location=recorded, simulated or none and provides by-location totals. Group counts are workspace counts before search/status filtering; result totals reflect those filters. On an older server without valid complete counts, the interface labels its fallback as limited coverage of the loaded page rather than claiming all located reports are shown.

Numbered pins match list entries. Overlapping coordinates are not moved to create artificial separation; the list keeps every report accessible. Invalid or unplottable rows remain reviewable without invented pins. The inspector shows the image, status, saved-box count, coordinates and source. Selection color is not severity, and draft priority is omitted from this inspector to avoid suggesting otherwise.

Coordinates and external links

For a new photo, Overview accepts optional latitude and longitude with Manual or Simulated demo provenance. The map does not collect GPS, extract location from photo/video metadata or edit saved coordinates. Zero coordinates are valid. Only valid manual/device-GPS positions offer an external Google Maps search, with an explicit notice that clicking shares coordinates with Google; no external map is loaded automatically and no such link is offered for simulation or unknown provenance.

Curated help not an LLM

Website guide is available in English and Arabic even before connection. It matches typed questions to curated instructions after language normalization; conflicting intents can fall back instead of guessing. It receives screen context and navigation callbacks, not report contents. Questions remain in component memory and are cleared when the panel closes.

Open this screen uses allowlisted destinations and existing navigation guards. It never starts a camera, uploads media, chooses a model, saves a review or changes permission. How this map works opens the map topic directly and returns focus to its button on close. Keyboard focus containment, Escape, RTL layout and screen highlights help users find the relevant controls. The guide distinguishes unavailable website report export from the approved-dataset ZIP available in Learning.

Appendix H Reviewed dataset contract

The learning loop is explicit: correct the image, save the review, approve the checked example, export a dataset, train and evaluate separately, then import and activate an accepted candidate. Saving a correction or approval does not change current weights. [R7, R17]

Approval contract

Confirmed, Assigned and Resolved reports can be eligible. Approval requires the current revision, permission to use the image, a capture-group ID and a train/validation/test assignment. Empty annotations need an additional reviewed-negative confirmation so a failed or uninspected empty prediction does not silently become a negative training example.

Approval is bound to the saved image, annotations, status and revision. Editing the report makes the approval stale; stale examples are excluded until checked and approved again. The workflow rejects an already-approved exact duplicate image and prevents one capture group from crossing splits. Export checks those conditions again and verifies the stored image hash.

Export component	Contents or boundary
Images and labels	Stored JPEGs and normalized YOLO box labels in train, val and test folders
data.yaml	Relative split paths and explicit class-name order
manifest.json	Image hash, revision, group, split, dimensions, annotations, reviewer and initial model identity
Capacity	Up to 1,000 examples and 512 MiB of image data per local package
Excluded metadata	No live tokens, GPS coordinates or review notes; images and reviewer identifiers remain private

Class mapping and split policy

Exported indices are 0 D00, 1 D10, 2 D20, 3 D40 and 4 OTHER. Some imported five-class checkpoints use OTHER at 3 and D40 at 4. Data must be remapped by class name before training; an index-preserving merge can swap potholes and OTHER while appearing syntactically valid.

Capture-group constraints only work if groups reflect real recording, route or damage identities. Exact hashes catch identical stored images, not transformed duplicates, nearby video frames or repeated views. The application does not automatically infer those relationships or create an independent test by random splitting.

Promotion must follow evaluation

A separate training environment should retain representative prior training data, protect validation/test groups and record the full recipe. Choose thresholds and checkpoints on validation, then evaluate a frozen candidate on untouched groups. Compare per-class errors and runtime under the same protocol, keep a rollback checkpoint and activate deliberately only after acceptance.

No automated retraining, live weight update or model promotion is implemented by approval/export. The presence of an export tool is not evidence that private reviewed reports were used in a completed training run. A download from Learning also does not substitute for the unavailable general PDF/CSV report-export control.

Evidence and references

Repository references below are pinned to the source snapshot used for the technical description. Dated experiment and release documents retain their historical scope; later provenance and verification records supersede stale counts or status statements, not the underlying original results.

[R1] CRDDC 2022 official data and submission guidance. Four target classes, unannotated official test release and challenge evaluation scope. Accessed 1 November 2026.
Official data page

[R2] RoadScan API and report persistence. Input limits, saved-report state, revision checks, location grouping, audit and mock boundaries.
api.py at the pinned revision

[R3] Team models and experiments. Supplied data narrative, E1–E5, Main thresholds, class metrics and the three-request timing diagnostic.
Team models and results

[R4] Model inventory and historical model report. Full hashes, exact bytes, roles, class/configuration evidence, baseline recovery and dated preflight results.
Machine readable model inventory

Historical model report

[R5] Inference adapter and reproducibility guide. Checkpoint hash checks, result states, class mapping, CPU inference and temporary raw-frame API.
Inference and reproducibility

[R6] Video and live implementation. Session limits, sampling, coverage, cancellation and frontend damage-sequence grouping.
video_analysis.py at the pinned revision

Live and models including the RT-DETR boundary regression

[R7] Review and training approval implementation. Strict annotations, revision-bound approval, capture groups, normalized export and privacy boundaries.
training_data.py at the pinned revision

[R8] Model voting implementation. Distinct checkpoint voting, IoU clustering, quorum, failures and worker lifetime isolation.
voting.py at the pinned revision

[R9] Map and guide clarity. Current location groups, independent paging, safe legacy fallback, coordinate privacy and stated limits.
Map and guide clarity

[R10] Website guide behavior and verification. Curated matching, allowed navigation, focus/RTL and conservative child-state guards.
Website guide

Related source links: CRDDC submissions https://crddc2022.sekilab.global/submissions/; current media frontend frontend/src/FullVideoAnalysis.tsx; historical model narrative Deliverables/Model-Report.md. Source commit: 58e7c3e494b5e1c5f2ca4d431b24d3f099003205.

Evidence records and reproducibility notes

[R11] Model organization and peer supplement. 22 verified identities, proposed aliases, two separately peer-reported weights and current publication accounting.
Model organization

[R12] Trainer provenance and release update. Verified current bytes, trainer-attested stock versions/lineage, ResNet initializer update and unresolved selection history.
Trainer provenance

[R13] RoadScan road package identity audit. Read-only reconstruction of 923 frames, 580/343 membership, selection reuse and cross-group/adjacent-frame evidence.
Fine tune source and evidence context

[R14] Ultralytics performance metrics. Definitions of IoU, AP, mAP50, mAP50–95, precision, recall and F1. Accessed 1 November 2026.
Official metric definitions

[R15] Current duplicate finder. DINOv2 crop preprocessing, cache, spatial/provenance filtering, bounds and unvalidated threshold.
duplicate_finder.py

[R16] Priority and dataset sensitivity. App score components and deterministic ranking-sensitivity mechanism.
priority.py and core formula

[R17] Learning and deployment guide. Review-to-export boundary, label order, split policy and separate training/promotion.
Learning and deployment

[R18] Installation and model import guidance. Documented setup and remaining clean-machine acceptance limits.
Run on another computer

[R19] Runtime persistence record. Dated 31 October source-image recreation and state-preservation evidence; not deployment of 1 November main.
Runtime persistence

[R20] Final approved merge verification receipt. 1 November 18:45 UTC record for exact main, matching tree, 257/3 backend, 42 frontend and integrated synthetic checks.
Public verification receipt

[R21] Combined candidate verification receipt. Integration paging, dependency audit with zero findings and documentation-link check; read with the final exact-main receipt.
Combined verification receipt

[R22] OLO AIEP Projects v2. Supplied official project brief, 43 pages; Track A on pp. 5–11, report/demo/artifact requirements and rubric on pp. 38–43. The report requirement is 10–15 main pages plus appendices. The team-collected 40–80-photo city set is test-only. The supplied brief, not the suggested role split, governs the requirement matrix.

[R23] Peer technical provenance receipt. Public source-backed record dated 1 November 2026: Main settings, fine-tune counts and separate P1/P2/P3 protocols. Recorded and locally executed statements remain peer evidence, not a new independent run.
Public training and protocol receipt

R13 audit identity: RoadScan_road_package.zip, 377,719,771 bytes, SHA256 7463144650f26bd50537ccdb1174a5933b2fdabcf2248f18b21129e0d777b2e7. The dated local audit and frame/label manifest preserve its findings. Supplied big_test_4_models.json SHA256: bd7fb7c71a5d8e8a5a942d016da1f428c07365d164d7d9af15a3cd5c2b25f153. Original package code and model files were not executed for that audit. Public context is linked above; the detailed package audit is not represented as a public repository artifact.